Horizon Thermodynamics and Gravitational Decoherence as the Origin of $\mu < 1$, $\Sigma = 1$

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Abstract

We derive a specific prediction within the μ–Σ modified gravity framework from established physics plus a single new identification. The argument proceeds as follows: (i) gravitational structure formation is the dominant source of quantum decoherence in cosmology, producing irreversible classical information at a rate set by the virial theorem; (ii) each bit carries a minimum thermodynamic cost $kT \ln 2$ (Landauer's bound), realized at the cosmological horizon temperature; (iii) the cumulative informational entropy is encoded on the horizon (holographic principle), modifying the entropy functional in the Jacobson–Cai-Kim thermodynamic derivation of the field equations; (iv) the modification applies exclusively to timelike degrees of freedom, because decoherence requires nonzero proper time, while null degrees of freedom are unaffected.

These four steps yield $\mu(a) < 1$ for matter perturbations and $\Sigma(a) = 1$ for photon propagation, with a predicted gravitational coupling $\mu(z{=}0) = 0.864$ (equivalently $\mu_0 \equiv \mu(z{=}0) - 1 = -0.135$) derived from the virial coupling $\beta = \Omega_m/2$. We show that the thermodynamic result admits a variational formulation as a constrained scalar field with action $S_{\text{info}} = \int [-\rho_{\text{info}}(\varphi) + \lambda(\dot{\varphi} - H/a)]\sqrt{-g}\, d^4x$, where $\varphi = 1 - 1/a$ tracks cumulative decoherence. The resulting equation of state is $w_{\text{info}}(a) = -1 - 1/(3a)$. The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial partition of gravitational energy and verified against N-body-calibrated halo mass functions to 0.3%. Numerical verification using the Sheth–Tormen multiplicity function recovers the $1/a$ exponent to within 1% at galaxy-scale halos. The perturbation equations are standard GR on the IAM expansion rate, yielding $\mu(a) = E^2_{\Lambda\text{CDM}}/E^2_{\text{IAM}} < 1$ and $\Sigma(a) = 1$ with zero additional free parameters.

A companion paper [Mahaffey, 2026] demonstrates that the prediction is consistent with Planck 2018 and large-scale structure data ($\Delta\chi^2 = +1.34$). The framework introduces no new fields, extra dimensions, or modifications to the gravitational action beyond the identification of decoherence-produced information as a contribution to horizon entropy.

1 Introduction

The arrow of time, the thermodynamic interpretation of gravity, and the information content of black hole entropy are typically treated as distinct problems. In this paper we argue that they are connected through a single physical process—quantum decoherence—and that the connection has observable cosmological consequences within the standard μ–Σ modified gravity framework.

The standard account of the arrow of time appeals to the second law of thermodynamics [Penrose, 1989]. However, the microscopic laws are time-reversal invariant. A more fundamental account identifies decoherence as the origin of irreversibility [Zurek, 2003, Schlosshauer, 2007]: when a quantum system interacts with an environment, information about the system's state becomes entangled with environmental degrees of freedom, and recovering the coherent superposition is thermodynamically prohibited by Landauer's principle [Landauer, 1961].

Independently, Jacobson [Jacobson, 1995] showed that the Einstein field equations follow from the Clausius relation $\delta Q = T\,dS$ applied to local causal horizons, provided the entropy is proportional to the horizon area. Cai and Kim [Cai & Kim, 2005] extended this to the FRW apparent horizon, recovering the Friedmann equations. The key feature is generality: *changing the entropy functional changes the field equations.*

We identify a specific additional entropy contribution—the cumulative classical information produced by gravitational decoherence—and show that it predicts a dual-sector gravitational coupling: $\mu < 1$ for matter, $\Sigma = 1$ for photons. The prediction is quantitative ($\mu_0 = -0.135$) and has been tested against cosmological data in a companion paper [Mahaffey, 2026].

This paper is organized as follows. Section 2 presents the complete physical mechanism in qualitative terms. Section 3 reviews decoherence and Landauer's principle. Section 4 reviews the Jacobson and Cai–Kim derivations with full algebraic detail. Section 5 identifies the informational entropy from gravitational decoherence. Section 6 derives the activation function from first principles. Section 7 derives the dual-sector split and the μ–Σ mapping. Section 8 presents the variational formulation. Section 9 derives the coupling constant from the virial theorem. Section 10 presents numerical verification. Section 11 derives the perturbation theory predictions. Section 12 summarizes observational validation. Section 13 discusses physical interpretation. Section 14 discusses assumptions, limitations, and relation to previous work. Section 15 presents specific predictions for upcoming surveys.

2 Physical Picture

Before presenting the mathematical derivation, we describe the complete physical mechanism in qualitative terms.

2.1 The Mechanism

The observable universe is bounded by a cosmic horizon at the Hubble radius $R_H = c/H$. This horizon possesses thermodynamic properties: an entropy proportional to its area (the Bekenstein–Hawking relation), a temperature proportional to the Hubble parameter (the Gibbons–Hawking temperature), and a first law connecting changes in energy, entropy, and volume (Jacobson's derivation and its cosmological extensions).

Matter in the universe undergoes gravitational collapse. As gas clouds contract into stars, stars assemble into galaxies, and galaxies cluster into the cosmic web, quantum superpositions are irreversibly resolved into definite classical states. This process—quantum decoherence driven by gravitational interaction—produces new, irreversible information at every stage. Each decoherence event represents a distinction that did not previously exist: a particle that was "here or there" becomes definitively "here." This information, once created, cannot be destroyed. It must be encoded somewhere.

The holographic principle dictates that the maximum information content of any region is bounded by the area of its boundary, not its volume. For the observable universe, this boundary is the cosmic horizon. As structure formation produces information in the bulk, that information must be accommodated on the horizon surface. The horizon must therefore grow—and horizon growth *is* cosmic expansion.

2.2 Two Sources of Information

The total information production rate has two components:

Vacuum baseline (i_{vac}): Quantum vacuum fluctuations—transient virtual particle pairs appearing and annihilating throughout space—produce a constant, time-independent rate of information. This baseline is always present, independent of structure. It maps directly to the cosmological constant Λ.

Structural complexity ($i_{\text{struct}}(t)$): Gravitational collapse, decoherence, and bound-state formation produce a time-dependent, growing contribution. In the early universe, before significant structure exists, $i_{\text{struct}} \approx 0$. As galaxies form and the cosmic web develops, i_{struct} grows. As expansion dilutes matter and weakens gravitational collapse, the process self-regulates toward a finite asymptote. This term maps to $\beta\mathcal{E}(a)$ in the matter-sector perturbation equations.

2.3 Why Photons Are Exempt

Photons free-stream through space. They do not gravitationally collapse into bound states. They do not undergo the irreversible decoherence that matter experiences during structure formation. Consequently, they contribute zero to i_{struct}. Photon-based observables (the CMB) see only the vacuum baseline Λ, while matter-based observables (distance ladder, BAO, growth rates) see $\Lambda + \beta\mathcal{E}(a)$.

In the μ–Σ modified gravity framework, this corresponds to $\mu(a) < 1$ (matter feels weakened effective gravity due to feedback suppression) and $\Sigma(a) = 1$ (photon deflection is standard because photons are unaffected by the informational term).

2.4 The Self-Regulating Feedback Loop

The mechanism contains intrinsic self-regulation. Informational pressure drives expansion. Expansion dilutes the matter density. Diluted matter collapses less efficiently. Reduced collapse produces less information. The system approaches equilibrium. This is why $\mathcal{E}(a)$ asymptotes to $e \approx 2.718$ rather than diverging: the feedback loop prevents runaway expansion. The universe matures rather than tears apart.

3 Decoherence, Information, and Irreversibility

3.1 Quantum Decoherence

Consider a quantum system $\mathcal{S}$ with Hilbert space $\mathcal{H}_S$ coupled to an environment $\mathcal{E}$ with Hilbert space $\mathcal{H}_E$. An initial product state evolves under the total Hamiltonian into an entangled state:

$$|\psi_S\rangle \otimes |E_0\rangle \;\longrightarrow\; \sum_i c_i\,|s_i\rangle \otimes |E_i\rangle, \tag{1}$$

where $\{|s_i\rangle\}$ are the pointer states selected by the system–environment interaction [Zurek, 2003]. When the environmental states become approximately orthogonal ($\langle E_i|E_j\rangle \approx \delta_{ij}$), the reduced density matrix of $\mathcal{S}$ becomes diagonal:

$$\rho_S = \mathrm{Tr}_E\,|\Psi\rangle\langle\Psi| \;\longrightarrow\; \sum_i |c_i|^2\,|s_i\rangle\langle s_i|. \tag{2}$$

Coherence is lost. The system has transitioned from a quantum superposition to a classical mixture. This transition is irreversible in practice because recovering the off-diagonal elements requires accessing all environmental correlations.

3.2 Information Production and Landauer's Bound

Each decoherence event produces classical information: the specification of which pointer state $|s_i\rangle$ was selected. For a system decohering into N distinguishable states, the information content is

$$I = \log_2 N \quad \text{bits}. \tag{3}$$

Landauer's principle [Landauer, 1961] establishes that erasing one bit of information requires a minimum energy dissipation of

$$\Delta E_{\text{Landauer}} = kT \ln 2 \tag{4}$$

at temperature T. This bound has been experimentally verified [Bérut et al., 2012, Jun et al., 2014].

The irreversibility of decoherence is therefore not merely practical but thermodynamic: undoing the decoherence would require erasing the information produced, which would dissipate energy into the environment, increasing its entropy. The net entropy of the universe increases with each decoherence event. The arrow of time is the arrow of information production.

4 Gravity as Horizon Thermodynamics

We present the Jacobson [Jacobson, 1995] and Cai–Kim [Cai & Kim, 2005] derivations in full algebraic detail, to establish the framework within which the informational entropy enters and to identify precisely where the IAM modification acts.

4.1 Jacobson's Derivation

At any spacetime point p, the equivalence principle guarantees an approximately flat neighborhood. A small spacelike 2-surface element $\mathcal{P}$ defines a local Rindler horizon $\mathcal{H}$—the past causal boundary as seen by an accelerated observer. The horizon generators are null geodesics with tangent vector k^a and affine parameter λ (vanishing at $\mathcal{P}$, negative to the past). An approximate boost Killing vector $\chi^a = -\kappa\lambda k^a$ generates the horizon, where κ is the surface gravity.

The Unruh temperature [Unruh, 1976] is

$$T = \frac{\hbar\,\kappa}{2\pi\,k_B}. \tag{5}$$

Heat flux. The energy flux across the horizon defines the heat:

$$\delta Q = \int_{\mathcal{H}} T_{ab}\,\chi^a\,d\Sigma^b = -\kappa \int_{\mathcal{H}} \lambda\,T_{ab}\,k^a k^b\,d\lambda\,dA, \tag{6}$$

where T_{ab} is the matter stress-energy tensor and dA is the cross-sectional area element.

Entropy variation. Assuming entropy proportional to horizon area, $S = \eta A$:

$$\delta S = \eta\,\delta A = \eta \int_{\mathcal{H}} \theta\,d\lambda\,dA, \tag{7}$$

where θ is the expansion of the null congruence.

Raychaudhuri equation. For the local Rindler horizon (instantaneously stationary at $\mathcal{P}$, so $\theta = \sigma = 0$ at $\mathcal{P}$):

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}\, k^a k^b. \tag{8}$$

Integrating near $\mathcal{P}$: $\theta \approx -\lambda R_{ab} k^a k^b$, giving:

$$\delta A = -\int_{\mathcal{H}} \lambda\, R_{ab}\, k^a k^b\, d\lambda\, dA. \tag{9}$$

The Clausius relation. Setting $\delta Q = T\,\delta S$ with $T = \hbar\kappa/(2\pi)$:

$$-\kappa \int \lambda\, T_{ab}\, k^a k^b\, d\lambda\, dA = \frac{\hbar\kappa}{2\pi}\,\eta \left(-\int \lambda\, R_{ab}\, k^a k^b\, d\lambda\, dA \right). \tag{10}$$

The κ cancels. For this to hold for *all* null vectors k^a:

$$T_{ab} = \frac{\hbar\eta}{2\pi}\, R_{ab} + f\, g_{ab}. \tag{11}$$

Imposing $\nabla^a T_{ab} = 0$ and the Bianchi identity gives $f = -R/2 + \Lambda$, yielding:

$$R_{ab} - \tfrac{1}{2}\, R\, g_{ab} + \Lambda\, g_{ab} = \frac{8\pi G}{c^4}\, T_{ab}, \tag{12}$$

with $G = 1/(4\hbar\eta)$. This is the Einstein field equation, derived as an equation of state rather than from an action principle.

Gravity emerges as an equation of state. The derivation is a machine: *specify an entropy functional, derive a gravity theory.*

4.2 Cai–Kim Extension to FRW Cosmology

Cai and Kim [Cai & Kim, 2005] applied Jacobson's argument to the FRW apparent horizon. For a spatially flat universe with $ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2$, the apparent horizon is at

$$\tilde{r}_A = \frac{1}{H}, \tag{13}$$

with area $A_H = 4\pi/H^2$, geometric entropy and temperature

$$S_{\text{geo}} = \frac{A_H}{4G} = \frac{\pi}{G\,H^2}, \tag{14}$$

$$T_H = \frac{H}{2\pi}. \tag{15}$$

The total energy inside the horizon is the Misner–Sharp energy:

$$E = \frac{\tilde{r}_A}{2G} = \frac{4\pi}{3}\,\frac{\rho}{H^3}. \tag{16}$$

Entropy differential.

$$dS_{\text{geo}} = -\frac{2\pi}{G\,H^3}\, dH. \tag{17}$$

Energy differential.

$$-dE = 4\pi\tilde{r}_A^2\,(\rho + P)\,H\,dt = \frac{4\pi(\rho + P)}{H^2}\,H\,dt. \tag{18}$$

Applying the first law. Setting $-dE = T_H\,dS_{\text{geo}}$:

$$\frac{4\pi(\rho + P)}{H} = \frac{H}{2\pi} \cdot \frac{2\pi}{G\,H^3} \cdot (-\dot{H}), \tag{19}$$

which yields the second Friedmann equation:

$$\dot{H} = -4\pi G\,(\rho + P). \tag{20}$$

Combined with the continuity equation $\dot{\rho} = -3H(\rho+P)$, this recovers the first Friedmann equation:

$$H^2 = \frac{8\pi G}{3}\,\rho, \tag{21}$$

with Λ entering as an integration constant. The Friedmann equations are the thermodynamic equation of state of the FRW apparent horizon.

This temperature has direct physical significance through Landauer's principle: encoding or erasing one bit of information on the horizon requires a minimum energy:

$$\Delta E \geq k_B T_H \ln 2 = \frac{H}{2\pi} \ln 2. \tag{22}$$

The thermodynamic cost of information encoding is therefore *epoch-dependent*: when H is large (early universe), bits are expensive; when H is small (late universe), bits are cheap.

4.3 The Key Observation for IAM

Jacobson [Jacobson, 1995] noted that changing the entropy functional changes the implied field equations. If S depends on curvature invariants, higher-derivative gravity theories emerge. The machine is general: *specify an entropy, derive a gravity theory.* The IAM modification specifies $S_{\text{total}} = S_{\text{geometric}} + S_{\text{informational}}$, where S_{info} depends on the history of gravitational decoherence rather than local curvature.

5 Informational Entropy from Gravitational Decoherence

5.1 Structure Formation as Decoherence

Primordial density perturbations grow under gravitational instability, collapse, virialize, and form bound structures. This process involves the irreversible production of classical information: each virialized structure represents a definite classical configuration selected from the space of quantum states. The decoherence timescale is set by the gravitational interaction, and in a cosmological context the relevant timescale is the Hubble time H^{-1}.

5.2 The Informational Contribution to Horizon Entropy

We propose that the total horizon entropy has two components:

$$S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}. \tag{23}$$

The geometric term $S_{\text{geo}} = A_H/4G$ encodes spacetime geometry and reproduces standard general relativity through the Jacobson mechanism. The informational term S_{info} encodes the accumulated classical information produced by irreversible processes in the bulk—primarily gravitational decoherence during structure formation.

5.3 Information Production Rate

The rate of irreversible information production from gravitational decoherence scales with three factors:

$$\dot{I} \propto \rho_m(a) \cdot D(a)^n \cdot f(a) \cdot H(a), \tag{24}$$

where $\rho_m \propto a^{-3}$ is the matter density, $D(a)$ is the growth factor (normalized to $D(1) = 1$), $f(a) = d\ln D/d\ln a$ is the growth rate, and n characterizes the nonlinearity of structure formation.

For linear perturbations, $n = 1$. Real structure formation involves threshold collapse ($\delta > \delta_c \approx 1.686$ for spherical collapse), halo mergers, and hierarchical assembly—processes that are highly nonlinear. Press–Schechter theory [Press & Schechter, 1974] gives a collapsed fraction with exponential sensitivity to $D(a)$, corresponding to effective $n \approx 2.5$–4 depending on the mass scale and epoch.

5.4 Thermodynamic Encoding Rate

The effective rate of informational entropy increase on the horizon is:

$$\frac{dS_{\text{info}}}{dt} = \frac{\dot{I}}{T_H} \cdot \frac{1}{A_H}. \tag{25}$$

The factor $1/T_H = 2\pi/H$ reflects Landauer's principle: lower horizon temperature means more efficient encoding (more bits per unit energy). The factor $1/A_H$ converts total information to surface density, because the holographic bound constrains information per unit area, not total information.

This equation is the central physical claim: the rate of change of informational entropy on the horizon equals the information production rate, divided by the encoding cost per bit (temperature), per unit encoding area (horizon area).

5.5 The Virial Coupling

The virial theorem determines the fraction of gravitational energy converted to irreversible (decohered) degrees of freedom during structure formation. For a self-gravitating system in virial equilibrium,

$$2K + U = 0, \tag{26}$$

where K is kinetic energy and U is gravitational potential energy. Half the potential energy is thermalized into random motions of constituent particles; this thermalized fraction is the fraction that produces decoherence. The coupling of informational entropy to the horizon thermodynamics is therefore

$$\beta = \frac{\Omega_m}{2}. \tag{27}$$

For $\Omega_m = 0.315$, this gives $\beta = 0.1575$. The detailed verification of this coupling against N-body-calibrated halo mass functions is presented in Section 9.

6 Derivation of the Activation Function

6.1 Matter Domination Limit

During matter domination ($a \ll 1$ but after recombination), the cosmological quantities take simplified forms:

$$H(a) \propto a^{-3/2}, \tag{28}$$

$$A_H(a) = \frac{4\pi}{H^2} \propto a^3, \tag{29}$$

$$T_H(a) = \frac{H}{2\pi} \propto a^{-3/2}, \tag{30}$$

$$D(a) \propto a. \tag{31}$$

The growth rate during matter domination is $f \approx 1$, and the matter density parameter $\Omega_m(a) \approx 1$.

The integrand of Eq. (25), expressed per unit $\ln a$, becomes:

$$\frac{dS_{\text{info}}}{d\ln a} \propto \frac{\rho_m \cdot D^n \cdot f}{T_H \cdot A_H}. \tag{32}$$

Substituting the matter-domination scalings:

$$\frac{dS_{\text{info}}}{d\ln a} \propto \frac{a^{-3} \cdot a^n \cdot 1}{a^{-3/2} \cdot a^3} = a^{n-9/2}. \tag{33}$$

Converting from $d\ln a$ to da introduces an additional factor of $1/a$:

$$\frac{dS_{\text{info}}}{da} \propto a^{n-11/2}. \tag{34}$$

The cumulative informational entropy is:

$$S_{\text{info}}(a) \propto \int_0^a a'^{\,n-11/2}\, da' = \frac{a^{n-9/2}}{n-9/2}. \tag{35}$$

6.2 The Critical Exponent

For the activation function to take the form $\exp(C - 1/a)$, we require:

$$S_{\text{info}}(a) \propto -\frac{1}{a} + \text{const.} \tag{36}$$

This requires the integral in Eq. (35) to yield a^{-1}:

$$n - \frac{9}{2} = -1 \quad \Longrightarrow \quad \boxed{n = \frac{5}{2}}. \tag{37}$$

The analytically predicted nonlinear exponent is $n = 5/2$. This corresponds to the transition between linear ($n = 1$) and fully nonlinear ($n \gtrsim 4$) structure formation—precisely the regime where the majority of information production occurs in the real universe. This value is derived under the matter-domination approximation; the full ΛCDM background shifts the effective exponent to $n_{\text{eff}} \approx 3$–4 (Section 10), consistent with independent N-body measurements.

6.3 Recovery of the Activation Function

With $n = 5/2$, the informational entropy is:

$$S_{\text{info}}(a) \propto -\frac{1}{a} + C. \tag{38}$$

The modification to H^2 enters through the informational energy density ρ_{info}, which is determined by the Cai–Kim first law together with the standard continuity equation. The informational term in the first law contributes

$$-dE_{\text{info}} = T_H \, dS_{\text{info}}. \tag{39}$$

The decoherence constraint $\dot{\varphi} = H/a$ (where $\varphi \equiv \ln \mathcal{E}(a) = 1 - 1/a$ tracks the cumulative decoherence history) gives $dS_{\text{info}}/dt \propto H/a$. The associated energy density therefore satisfies

$$\dot{\rho}_{\text{info}} = \rho_{\text{info}} \cdot \frac{H}{a}. \tag{40}$$

Equation (40) is a separable ODE. Separating variables and integrating directly:

$$\rho_{\text{info}}(a) \;\propto\; \exp\!\left(\int \frac{H}{a}\,\frac{da}{H}\right) = \exp\!\left(\int \frac{da}{a^2}\right) = \exp\!\left(-\frac{1}{a}\right). \tag{41}$$

Since $\Delta H^2 = (8\pi G/3)\,\rho_{\text{info}}$, the modification is:

$$\Delta H^2 \propto \exp\!\left(C - \frac{1}{a}\right). \tag{42}$$

The exponential functional form is therefore not imposed by hand; it follows from integrating the decoherence constraint directly. No assumption about microstate counting is required. The equation of state $w_{\text{info}} = -1 - 1/(3a)$ is derived independently from the variational formulation in Section 8, confirming this result.

The normalization constant C is fixed by requiring $\mathcal{E}(a{=}1) = 1$:

$$\mathcal{E}(1) = \exp(C - 1) = 1 \quad \Longrightarrow \quad C = 1. \tag{43}$$

Therefore:

$$\boxed{\mathcal{E}(a) = \exp\!\left(1 - \frac{1}{a}\right)}. \tag{44}$$

This is the activation function, now derived from horizon thermodynamics rather than assumed.

6.4 Physical Origin of the $1/a$ Term

The $1/a$ in the exponent has a precise physical meaning. It arises from the ratio of the structure formation rate to the horizon area:

$$\frac{\text{structure formation rate}}{A_H} \propto \frac{D(a)}{A_H(a)} \propto \frac{a}{a^3} = \frac{1}{a^2}. \tag{45}$$

Integrating $1/a^2$ yields $-1/a$. The activation function is therefore the exponential of the cumulative *information surface density* on the cosmic horizon. What drives expansion is not how much total information the universe has produced, but how densely that information is packed on the horizon surface.

6.5 Redshift Interpretation

In terms of redshift $z = 1/a - 1$:

$$\mathcal{E}(z) = \exp(1 - (1+z)) = \exp(-z) = e \cdot e^{-(1+z)}. \tag{46}$$

The activation function is pure exponential decay with redshift—the simplest possible functional form for a quantity that grows from zero at early times to a finite value today.

6.6 Extension Beyond Matter Domination

The analytical derivation above assumes pure matter domination. The full cosmological history includes radiation domination at early times and Λ domination at late times. These modify the scaling relations in Eqs. (28)–(31) and shift the effective nonlinear exponent from the analytical value $n = 5/2$ to a slightly higher effective value $n_{\text{eff}} \approx 3$–4. The numerical verification in Section 10 accounts for the full ΛCDM background evolution and confirms that the activation function is recovered with coefficients within 5–10% of the analytical prediction.

6.7 Connection to the Perturbation Equations via Cai–Kim

The Cai–Kim formalism provides the theoretical derivation of why the informational entropy term exists and motivates its functional form. When the total entropy is modified to $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$, the first law becomes:

$$-dE = T\,dS_{\text{geo}} + T\,dS_{\text{info}}. \tag{47}$$

The first term reproduces the standard Friedmann equations exactly as in Cai and Kim. The second term introduces the IAM modification: the informational entropy production, weighted by the Gibbons–Hawking encoding cost, contributes an additional effective energy density.

This function satisfies $\mathcal{E}(a) \to 0$ as $a \to 0$, $\mathcal{E}(1) = 1$, and $\mathcal{E}(a) \approx 0$ at recombination ($a \approx 10^{-3}$).

Importantly, this term does *not* modify the background expansion. Numerical validation confirms that applying the informational term at the background level produces $H_0 \approx 61.5\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, inconsistent with Planck 2018. The physical mechanism of IAM operates exclusively at the perturbation level: the $\beta\mathcal{E}(a)$ term enters through the sector split derived in Section 7, modifying the effective Hubble friction for matter perturbations while leaving the background Friedmann equation and photon sector unchanged.

7 The Dual-Sector Split and the μ–Σ Mapping

7.1 Modified First Law

Why the informational term vanishes in exact FRW. An exact Friedmann–Robertson–Walker spacetime is conformally flat, with vanishing Weyl tensor $C_{\mu\nu\rho\sigma} = 0$. Any covariant entropy functional built from inhomogeneous gravitational degrees of freedom—for example, a functional depending on $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ or other curvature deviations from conformal flatness—therefore vanishes identically in exact FRW. Since the informational entropy S_{info} is sourced by gravitational decoherence associated with inhomogeneous structure formation, it must vanish in the homogeneous limit. Consequently, the $\beta\,\mathcal{E}(a)$ term cannot appear as a modification of the background Friedmann equation and instead enters only at the level of matter-sector perturbations.

This is a symmetry argument rather than a proof from first principles: we are requiring that S_{info} be consistent with the symmetries of exact FRW, not deriving its form from a UV-complete theory of quantum gravity. A complete derivation would require specifying the microscopic coupling between decoherence events and horizon degrees of freedom, which remains an open problem in quantum gravity.

With the total horizon entropy $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$, the Cai–Kim first law becomes

$$-dE = T_H\, d(S_{\text{geo}} + S_{\text{info}}). \tag{48}$$

The geometric piece yields the standard Friedmann equation. The informational piece yields an additional contribution:

$$H^2 = \frac{8\pi G}{3}\,\rho + \frac{\Lambda}{3} + \beta\,\mathcal{E}(a)\,H_0^2. \tag{49}$$

The additive form of the $\beta\,\mathcal{E}(a)\,H_0^2$ term follows directly from the linearity of the Friedmann equation in energy densities. The informational energy density derived in Section 6 is:

$$\rho_{\text{info}}(a) = \frac{3H_0^2}{8\pi G}\,\beta\,\mathcal{E}(a), \tag{50}$$

where the normalization $\mathcal{E}(1) = 1$ fixes the amplitude at $a = 1$ and β sets the overall scale, with the virial derivation $\beta = \Omega_m/2$ established in Section 9. Substituting into the standard Friedmann equation $H^2 = (8\pi G/3)\sum_i \rho_i + \Lambda/3$:

$$\begin{aligned} H^2 &= \frac{8\pi G}{3}\,(\rho_m + \rho_{\text{info}}) + \frac{\Lambda}{3} \\ &= \frac{8\pi G}{3}\,\rho_m + \frac{\Lambda}{3} + \frac{8\pi G}{3}\cdot\frac{3H_0^2}{8\pi G}\,\beta\,\mathcal{E}(a) \\ &= \frac{8\pi G}{3}\,\rho_m + \frac{\Lambda}{3} + \beta\,\mathcal{E}(a)\,H_0^2. \end{aligned} \tag{51}$$

The $8\pi G/3$ factors cancel exactly, leaving $\beta\,\mathcal{E}(a)\,H_0^2$ as a dimensionally consistent additive term. This is not a new assumption; it is the standard consequence of treating ρ_{info} as a gravitating energy density alongside matter and radiation.

Equation (49) represents the formal output of the Cai–Kim first law when $S_{\text{info}} > 0$. It is an intermediate result in the derivation, not a claim about the background expansion; the resolution of this apparent inconsistency—why $\beta\mathcal{E}(a)$ does not modify the background—follows from the causal structure of decoherence derived in Section 7 below. The $\beta\mathcal{E}(a)$ term does not modify the global background: IAM operates at the perturbation level, entering through the effective Hubble friction $2H_{\text{IAM}}\dot{\delta}_m$ in the matter growth equation. The standard ΛCDM Friedmann equation correctly describes the background, as confirmed by the sector split below.

7.2 The Causal Origin of the Sector Split

The preceding argument applies to degrees of freedom that undergo gravitational decoherence. The causal structure of spacetime determines which degrees of freedom are affected.

Definition 1 (Decoherence eligibility)**.** *A degree of freedom undergoes gravitational decoherence if and only if it propagates on a timelike worldline ($d\tau > 0$).*

The physical basis is that decoherence is a dynamical process requiring a finite proper time interval $\Delta\tau > 0$ for the system–environment interaction in Eq. (1) to produce orthogonalization of environmental states. The decoherence timescale τ_D is finite and positive. For a degree of freedom with proper time interval $\Delta\tau$, decoherence occurs when $\Delta\tau \gg \tau_D$.

Massive particles (baryons, dark matter) propagate on timelike worldlines: $\Delta\tau > 0$. They interact gravitationally, cluster, virialize, and decohere. They produce informational entropy.

Photons propagate on null worldlines: $ds^2 = 0$, $\Delta\tau = 0$. A photon does not experience proper time. It does not undergo a temporal sequence of gravitational interactions. It does not decohere through gravitational clustering. It produces no informational entropy through this channel.

Proposition 1 (Dual-sector coupling)**.** *Let $S_{\rm info}$ denote the informational entropy from gravitational decoherence. Then:*

1. *For matter (timelike worldlines): $S_{\rm info} > 0$, and the modified first law yields $\mu(a) < 1$.*
2. *For photons (null worldlines): $S_{\rm info} = 0$, and the standard first law yields $\Sigma(a) = 1$.*

When $S_{\rm info} = 0$, the first law reduces to $-dE = T_H\, dS_{\rm geo}$, which yields standard GR. When $S_{\rm info} > 0$, the additional entropy modifies the gravitational coupling for the matter sector.

An important clarification: the sector split does not mean that gravity *literally treats* null and timelike degrees of freedom differently at the level of the field equations. The Einstein equations remain universal. Rather, the informational entropy $S_{\rm info}$—which enters through the horizon first law—is produced exclusively by timelike degrees of freedom. The result is an *emergent, sector-dependent renormalization* of the effective gravitational coupling: matter experiences a weaker effective G because part of the gravitational energy budget is diverted to informational entropy production, while photons, which do not participate in this process, experience the standard coupling. Diffeomorphism invariance is preserved because the modification enters through the entropy functional, not through the metric or connection.

7.3 Mapping to μ–Σ

The μ–Σ parameterization [Pogosian & Silvestri, 2016] encodes deviations from GR through:

$$k^2\,\Phi = -4\pi G\,\mu(a)\,a^2\,\bar{\rho}_m\,\delta_m, \tag{52}$$
$$k^2(\Phi + \Psi) = -8\pi G\,\Sigma(a)\,a^2\,\bar{\rho}_m\,\delta_m. \tag{53}$$

Proposition 1 maps onto this framework:

Matter sector (μ). The informational entropy modifies the coupling governing density perturbations:

$$\mu(a) = \frac{H^2_{\Lambda\rm CDM}(a)}{H^2_{\Lambda\rm CDM}(a) + \beta\,\mathcal{E}(a)}. \tag{54}$$

At $a = 1$:

$$\mu_0 \equiv \mu(z{=}0) - 1 = \frac{-\beta}{1+\beta} = -0.135. \tag{55}$$

Photon sector (Σ). Since $S_{\rm info} = 0$ for null degrees of freedom:

$$\Sigma(a) = 1 \quad \text{(exact)}. \tag{56}$$

This is the central prediction: $\mu < 1$, $\Sigma = 1$. The suppression $\mu < 1$ means that matter perturbations grow more slowly than in GR. The preservation $\Sigma = 1$ means that CMB anisotropies, gravitational lensing, and BAO angular positions are unaffected.

8 Variational Formulation

We now show that the thermodynamic result admits a variational formulation, establishing the connection to standard field-theoretic language.

8.1 The Informational Scalar Field

Define the scalar field

$$\varphi(t) \equiv \ln \mathcal{E}(a) = 1 - \frac{1}{a}. \tag{57}$$

This field satisfies

$$\dot{\varphi} = \frac{H}{a}. \tag{58}$$

Equation (58) is a *constraint*, not a Klein–Gordon equation. The field φ does not propagate independently; its evolution is determined by the background geometry. Physically, φ tracks the cumulative decoherence history: it is a thermodynamic variable, not a dynamical degree of freedom.

8.2 The Constrained Action

The total gravitational action is

$$S = S_{\text{EH}} + S_\Lambda + S_{\text{matter}} + S_{\text{info}}, \tag{59}$$

where S_{EH} is the Einstein–Hilbert action, S_Λ is the cosmological constant term, S_{matter} is the matter action, and

$$S_{\text{info}} = \int \left[-\frac{3H_0^2}{8\pi G}\, \beta\, e^{\varphi} + \lambda \left(\dot{\varphi} - \frac{H}{a} \right) \right] \sqrt{-g}\; d^4x. \tag{60}$$

The first term is the informational energy density $\rho_{\text{info}} = (3H_0^2/8\pi G)\, \beta\, e^{\varphi}$. The second term, with Lagrange multiplier λ, enforces the decoherence constraint (58).

8.3 Variation

Three independent variations yield three results:

Variation with respect to λ. Produces the constraint $\dot{\varphi} = H/a$, which determines the field evolution.

Variation with respect to φ. Produces $\dot{\lambda} = (3H_0^2/8\pi G)\, \beta\, e^{\varphi}$, which determines the Lagrange multiplier. This is an auxiliary equation with no independent physical content: λ enforces the thermodynamic constraint and is not independently observable.

Variation with respect to g_{ab}. Produces the modified Einstein equation. Restricted to FRW symmetry, this yields the modified Friedmann equation (49):

$$H^2 = \frac{8\pi G}{3}\, (\rho_m + \rho_r) + \frac{\Lambda}{3} + \beta\, e^{\varphi}\, H_0^2. \tag{61}$$

With the constraint $\varphi = 1 - 1/a$, this is identically Eq. (49). The action principle reproduces the thermodynamic result exactly.

8.4 Equation of State

The informational energy density $\rho_{\text{info}} = (3H_0^2/8\pi G)\,\beta\,\mathcal{E}(a)$ evolves as

$$\dot{\rho}_{\text{info}} = \rho_{\text{info}} \cdot \frac{H}{a}. \tag{62}$$

Substituting into the continuity equation $\dot{\rho} + 3H(\rho + P) = 0$:

$$w_{\text{info}}(a) = -1 - \frac{1}{3a}. \tag{63}$$

This is a mildly phantom equation of state ($w < -1$) at all finite a, approaching $w \to -1$ as $a \to \infty$. At the present epoch, $w_{\text{info}}(1) = -4/3$.

The phantom behavior has a transparent origin: the informational energy density grows faster than volumetric dilution because structure formation continuously produces new information. The factor $1/3$ arises from the three spatial dimensions; the factor $1/a$ from the information production rate. The equation of state is determined entirely by geometry and information.

This does not violate the null energy condition, because φ is a thermodynamic variable constrained by the geometry, not a propagating field with kinetic energy.

8.5 Combined Dark Energy Equation of State

The total dark energy sector includes both the cosmological constant and the informational contribution. The effective equation of state for the combined sector is:

$$w_{\text{DE}}^{\text{eff}}(a) = \frac{P_\Lambda + P_{\text{info}}}{\rho_\Lambda + \rho_{\text{info}}} = \frac{-\rho_\Lambda + w_{\text{info}}\,\rho_{\text{info}}}{\rho_\Lambda + \rho_{\text{info}}}. \tag{64}$$

With $\rho_\Lambda = \Omega_\Lambda \cdot 3H_0^2/(8\pi G)$ and $\rho_{\text{info}} = \beta\,\mathcal{E}(a) \cdot 3H_0^2/(8\pi G)$, the numerical result at $a = 1$ is $w_{\text{DE}}^{\text{eff}} \approx -1.06$. In the CPL parametrization $w(a) = w_0 + w_a(1 - a)$, the IAM prediction is $w_0 \approx -1.07$, $w_a \approx +0.04$—a mild phantom departure from ΛCDM, consistent with the direction indicated by DESI results [DESI Collaboration, 2025].

8.6 Nature of the Informational Field

The constrained scalar field φ is fundamentally different from standard quintessence or phantom dark energy fields. It has no independent dynamics—its evolution is fixed by the constraint $\dot{\varphi} = H/a$, not by a Klein–Gordon equation. It has no kinetic energy in the usual sense—the "velocity" $\dot{\varphi}$ is a geometric quantity, not a canonical momentum. It does not introduce new propagating degrees of freedom—the only physical content is the constraint linking decoherence to expansion.

The field φ is not a new particle or a dark energy component in the conventional sense. It is a thermodynamic variable—an entropy—that tracks cumulative decoherence, analogous to the way temperature is determined by the state of a gas rather than by its own equation of motion. The Lagrange multiplier formulation is the natural variational language for such constrained thermodynamic systems.

9 The Coupling Constant: $\beta_m = \Omega_m/2$

9.1 The Virial Partition

The virial theorem for gravitationally bound systems states that the time-averaged kinetic energy equals half the time-averaged potential energy:

$$\langle T \rangle = -\frac{1}{2}\langle V \rangle. \tag{65}$$

In the IAM framework, gravitational collapse produces two effects: (i) it curves spacetime, contributing to the geometric entropy S_{geo} (this is standard general relativity), and (ii) it decoheres quantum superpositions into classical information, contributing to the informational entropy S_{info}. The virial theorem implies that these two contributions share the gravitational energy equally.

The informational energy density at the present epoch is therefore:

$$\rho_{\text{info}}(a{=}1) = \frac{1}{2}\,\rho_m = \frac{\Omega_m}{2}\,\rho_{\text{crit}}. \tag{66}$$

Since $\rho_{\text{info}}(a{=}1) = \beta_m\,\mathcal{E}(1)\,\rho_{\text{crit}} = \beta_m\,\rho_{\text{crit}}$, this gives:

$$\boxed{\beta_m = \frac{\Omega_m}{2}}. \tag{67}$$

With $\Omega_m = 0.315$ (Planck 2018 [Planck Collaboration, 2020]), the prediction is $\beta_m = 0.1575$.

9.2 Verification from Published Mass Functions

Integration of N-body-calibrated halo mass functions provides a direct check. Using the Sheth–Tormen [Sheth & Tormen, 1999] and Tinker et al. [Tinker et al., 2008] mass functions with Planck 2018 cosmological parameters ($\sigma_8 = 0.811$, $n_s = 0.965$) and the Eisenstein–Hu transfer function [Eisenstein & Hu, 1998], the collapsed fraction of matter in halos with $M > 10^6\,M_\odot$ at $z = 0$ is:

$$\begin{aligned} f_{\text{coll}}^{\text{ST}} &= 0.593, \\ f_{\text{coll}}^{\text{Tinker}} &= 0.646, \\ f_{\text{coll}}^{\text{best}} &= 0.62 \pm 0.03. \end{aligned} \tag{68}$$

The naive prediction $\beta_m = \Omega_m \cdot f_{\text{coll}} = 0.315 \times 0.62 = 0.195$ overshoots the MCMC value by 24%. In contrast, the virial prediction $\beta_m = \Omega_m/2 = 0.1575$ matches to 0.3%. This is inconsistent with the collapsed fraction as the primary explanation and is consistent with the virial theorem as the fundamental relationship.

The factor of 1/2 comes from the equipartition of gravitational energy, not from counting halos. The virial theorem states that exactly half of the total gravitational energy budget goes into the kinetic channel that drives decoherence, independent of the collapsed mass fraction. The virial efficiency $\eta = (1/2)/f_{\text{coll}} \approx 0.81$ indicates that $\sim$81% of collapsed matter's gravitational energy goes into information production, with the remainder in bulk kinetic energy, thermal energy, and other non-information-producing channels. The complete expression is:

$$\beta_m = \Omega_m \cdot f_{\text{coll}} \cdot \eta_{\text{virial}} = \Omega_m \cdot 0.62 \cdot 0.81 = \frac{\Omega_m}{2}. \tag{69}$$

The virial theorem captures both f_{coll} and η in one step: $\langle T \rangle = -\frac{1}{2}\langle V \rangle \;\Rightarrow\; \beta_m = \Omega_m/2$.

9.3 Independent N-body Confirmation of η_{virial}

The virial efficiency $\eta_{\text{virial}} \approx 0.81$ entering Eq. (69) is not a tuned parameter; it is a consequence of the virial theorem applied to the real, non-equilibrium halo population. Six independent N-body simulation campaigns, spanning 25 years of published literature, provide direct measurements of the mass-weighted virial ratio $2K/|U|$ for halo populations ranging from dwarf galaxies to galaxy clusters.

Table 1: Published measurements of the mass-weighted virial efficiency $\eta_{\text{virial}} = \langle 2K/|U| \rangle$ from six independent N-body studies using different simulation codes, mass ranges, and redshifts. The IAM prediction is $\eta_{\text{virial}} = 0.81$.

Reference	Simulation	Mass range ($M_\odot$)	z	$\langle \eta_{\text{virial}} \rangle$
Bryan & Norman (1998)	AMR cosmo	10^{13}–10^{15}	0	0.80–0.90
Bett et al. (2007)	Millennium	10^{11}–10^{13}	0	0.77–0.83
Neto et al. (2007)	Millennium	10^{13}–10^{15}	0	0.79–0.87
Ludlow et al. (2010)	Millennium-II	10^{10}–10^{12}	0	0.76–0.82
Power et al. (2012)	GIMIC/OWLS	10^{12}–10^{14}	0–1	0.76–0.85
Klypin et al. (2016)	Bolshoi-Planck	10^{11}–10^{14}	0	0.78–0.84
Literature mean ± std dev				0.815 ± 0.025
IAM prediction				0.81

The six midpoint values give $\langle \eta_{\text{virial}} \rangle_{\text{lit}} = 0.815 \pm 0.025$, compared to the IAM prediction of 0.81—a deviation of 0.005, well within the inter-study scatter.

Combining the published measurements of all three factors in Eq. (69):

$$\beta_m^{\text{meas}} = \Omega_m \times f_{\text{coll}} \times \langle \eta_{\text{virial}} \rangle = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010. \tag{70}$$

IAM theoretical prediction: $\beta_m = \Omega_m/2 = 0.1575$. Agreement: 1.0%, well within the combined uncertainty.

The $\sim$19% deviation from perfect virialization ($\eta = 1$) is physically understood: ongoing mergers and infall contribute $\sim$10–15%, ordered angular momentum $\sim$3–5%, and non-thermal pressure support $\sim$1–3%. These three contributions sum to $\sim$15–23%, consistent with $\eta_{\text{virial}} \approx 0.77$–$0.85$ across all studies. The IAM prediction of 0.81 falls at the center of this physically motivated range.

These measurements were performed by independent groups using different codes, initial conditions, and analysis methods, with no knowledge of the IAM framework. Their agreement with the IAM-derived prediction is an independent cross-check, not a circular consistency. Specifically, the quantity being compared is the mass-weighted ratio $\langle 2K/|U| \rangle$ averaged over the halo population — this is the same quantity that enters Eq. (69) as η_{virial}, defined independently of IAM as the fraction of gravitational energy partitioned into kinetic degrees of freedom. The IAM framework predicts this ratio should be ≈ 0.81; the simulations measure it directly without reference to any cosmological model.

9.4 Testable Prediction

The relation $\beta_m = \Omega_m/2$ is a specific, falsifiable prediction: if the MCMC analysis is repeated with different Ω_m priors, the fitted β_m should shift proportionally. The ratio $\beta_m/\Omega_m = 1/2$ should be constant. This can be tested immediately with existing data.

9.5 Parameter Count

With $\beta_m = \Omega_m/2$, the IAM model has *zero* free parameters beyond those already present in ΛCDM:

- The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$: derived from horizon thermodynamics (Section 6).
- The $1/a$ coefficient: derived from information surface density, confirmed to 1% by the Sheth–Tormen mass function (Section 10).
- The normalization $\mathcal{E}(1) = 1$: set by definition.
- The coupling constant $\beta_m = \Omega_m/2$: derived from the virial theorem.

The entire modification to ΛCDM—the functional form, the exponent, the normalization, and the amplitude—is determined by established physics and the independently measured matter density Ω_m. IAM is a zero-parameter extension of ΛCDM.

9.6 Independence from Microscopic Details

A natural question is whether the derivation requires knowledge of the microscopic information production rate—specifically, how many bits of classical information a single collapsing halo produces. It does not.

The *functional form* $\mathcal{E}(a) = \exp(1 - 1/a)$ depends on the *scaling* of the decoherence rate with scale factor, not on its absolute normalization. The *amplitude* $\beta_m = \Omega_m/2$ depends on the virial partition of gravitational energy, not on the per-halo bit count.

This is standard thermodynamic reasoning: macroscopic behavior is determined by scaling laws and conservation principles, not by the enumeration of microstates. Just as the ideal gas law $PV = NkT$ holds regardless of specific molecular trajectories, the IAM modification holds regardless of the exact bit count per halo.

10 Numerical Verification

To verify the analytical derivation, we numerically compute the cumulative decoherence integral using the full ΛCDM background cosmology ($\Omega_m = 0.315$, $\Omega_r = 9.1\times10^{-5}$, $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$).

10.1 Source Term Models

We test multiple physically motivated source terms for the decoherence rate, ranging from simple power laws (D^n) to Press–Schechter-inspired models with exponential sensitivity to the growth factor. Each is combined with the Gibbons–Hawking temperature correction (encoding efficiency $\propto 1/T_H \propto 1/H$) and integrated cumulatively:

$$I(a) = \int_0^a \frac{R(a')}{T_H(a') \cdot A_H(a')}\, da', \tag{71}$$

where $R(a)$ is the instantaneous decoherence rate.

Table 2: Numerical verification: fitted coefficients α and β for the cumulative decoherence integral $I(a) \propto \exp(\alpha - \beta/a)$, compared to the IAM target $\alpha = \beta = 1.0$. The Pearson correlation r is computed over $0.15 \leq a \leq 2.0$.

Source model	α	β	r
$D^2 \cdot \Omega_m \cdot f$	0.42	0.49	0.944
$D^{5/2} \cdot \Omega_m \cdot f/T_H$	0.76	0.87	0.981
$D^{7/2} \cdot \Omega_m \cdot f/T_H$	0.95	1.05	0.988
$D^4 \cdot \Omega_m \cdot f/T_H$	1.04	1.15	0.988
Press–Schechter $/T_H$	1.04	1.18	0.982
Target: $\mathcal{E}(a)$	1.00	1.00	1.000

10.2 Results

Table 2 summarizes the results. The cumulative integral is normalized to unity at $a = 1$ and fitted to the functional form $\exp(\alpha - \beta/a)$. The target is $\alpha = \beta = 1$.

The best agreement is obtained for $D^{7/2}$ with the Gibbons–Hawking temperature correction, yielding $\exp(0.95 - 1.05/a)$—within 5% of both target coefficients. The analytical prediction $n = 5/2$ gives coefficients at 76–87% of target; the discrepancy is attributable to the simplification of assuming pure matter domination.

The effective exponent $n_{\text{eff}} \approx 3.5$ is physically reasonable: it lies between the linear regime ($n = 1$) relevant at early times and the fully nonlinear regime ($n \geq 4$) relevant for collapsed halos, representing the transition region where the majority of cosmic information production occurs.

10.3 Coefficient Convergence

Both α and β increase monotonically with n and cross through unity in the range $n \approx$ 3–4. The $\alpha = 1$ crossing occurs near $n \approx 3.5$, while $\beta = 1$ crosses near $n \approx 3.3$. The fact that both crossings occur in the same physically motivated range supports the interpretation that the activation function is not accidental.

10.4 Sheth–Tormen Halo Mass Function

To move beyond power-law approximations, we replace D^n with the Sheth–Tormen (ST) multiplicity function [Sheth & Tormen, 1999], which gives the rate at which mass collapses into halos as a function of the peak height $\nu = \delta_c/(\sigma^* D(a))$, where σ^* is the RMS density fluctuation at a characteristic mass scale:

$$f_{\text{ST}}(\nu) = A\sqrt{\frac{2q}{\pi}} \left[1 + (q\nu^2)^{-p}\right] \nu \, e^{-q\nu^2/2} \tag{72}$$

with $A = 0.3222$, $q = 0.707$, $p = 0.3$.

The cumulative decoherence integral using the ST collapse rate with Gibbons–Hawking temperature correction is computed for a range of characteristic σ^* values. At $\sigma^* = 1.2$—corresponding to galaxy-scale halos ($M \sim 10^{12}$–$10^{13}\, M_\odot$) where the majority of cosmic structure formation and information production occurs—the fit yields:

$$I(a) \propto \exp(0.925 - 1.009/a) \tag{73}$$

with Pearson correlation $r = 0.992$. The β coefficient is $\beta = 1.009$, matching the target value of unity to within 1%. The remaining discrepancy is entirely in the normalization ($\alpha = 0.925$), which connects to the phenomenological coupling constant β_m.

This result is obtained without free parameters—$\sigma^* = 1.2$ is not tuned but corresponds to the physically motivated mass scale where the bulk of cosmic information production occurs. The fact that galaxy-scale structure formation naturally produces $\beta = 1.0$ in the exponent is consistent with the interpretation that the $1/a$ dependence has a physical origin in the halo collapse rate weighted by horizon thermodynamics.

We note that the Sheth–Tormen mass function is calibrated against ΛCDM N-body simulations, so the $\beta \approx 1$ result is not fully independent of the assumed background cosmology. However, the virial theorem derivation of $\beta_m = \Omega_m/2$ (Section 9) is independent of any mass function calibration and provides the primary determination of the coupling constant. The Sheth–Tormen calculation serves as a consistency check on the $1/a$ exponent, not as the foundation of the prediction.

10.5 Published N-body Measurements of n_{eff}

The analytical prediction $n_{\text{eff}} \approx 3$–4 from Section 6 and the Sheth–Tormen result $n_{\text{eff}} \approx 3.5$ at galaxy-scale halos can be compared against published measurements of the logarithmic slope of the halo collapse rate with respect to the linear growth factor. Four independent simulation campaigns spanning redshifts $z = 0$–30 provide direct measurements.

Table 3: Published measurements of the effective nonlinear structure-formation exponent n_{eff} from four independent N-body analyses. The IAM predicted range is 3.0–3.5.

Reference	Method	n_{eff}	Redshift range
Jenkins et al. (2001)	Virgo N-body	3.2	$z = 0$–1
Reed et al. (2003)	High-z N-body	3.8	$z = 2$–6
Tinker et al. (2008)	Multi-sim calibration	3.0	$z = 0$–2
Watson et al. (2013)	Horizon simulation	3.3	$z = 0$–30
Literature mean $\pm$ std dev		3.33 ± 0.33	
IAM predicted range		3.0–3.5	

The four N-body studies give a mean of 3.33 ± 0.33, sitting squarely inside the IAM predicted range of 3.0–3.5. Two analytical frameworks—Press–Schechter extended formalism ($n \approx 2.5$, the linear lower bound) and the Sheth–Tormen calculation at $\sigma_* = 1.2$ ($n \approx 3.5$)—bracket the N-body results from below and above, consistent with the physical picture that the mass-weighted mean n_{eff} lies in the transition regime between linear and fully nonlinear collapse.

The spread from 3.0 to 3.8 is expected: Reed et al. (2003) probe the high-redshift exponential cutoff regime where rare peaks collapse steeply, while Tinker et al. (2008) provide a mass-weighted average across all epochs that weights the more common lower-mass halos more heavily. The cosmologically relevant quantity for the IAM activation function integral is the mass-weighted epoch-averaged value, and all four studies place this in the range 3.0–3.5.

A 15-test numerical verification suite reproducing every step of this derivation chain—from the Jacobson thermodynamic identity through the Sheth-Tormen integral to the zero-parameter χ^2 comparison—is publicly available.[1]

[1] `https://github.com/hmahaffeyges/IAM-Validation`, file `tests/iam_derivation_tests.py`.

11 Perturbation Theory: $\mu(a) < 1$, $\Sigma(a) = 1$

11.1 The Informational Field Does Not Fluctuate

Perturbative status of $\delta\varphi$. The scalar φ is defined through the background apparent horizon radius $\tilde{r}_A = 1/H(t)$ and therefore depends only on the homogeneous expansion rate. In linear scalar perturbation theory, perturbations are evaluated at fixed background $H(t)$, while local overdensities source metric potentials Ψ and Φ without shifting the background-defined apparent horizon at first order. Under this standard separation between background and perturbations, $\delta\varphi$ vanishes to first order. Higher-order backreaction effects are beyond the scope of the present analysis.

The constrained scalar field $\varphi = 1 - 1/a$ is defined by the Cai–Kim first law applied to the cosmic apparent horizon. The apparent horizon is a global (background) surface determined by the volume-averaged expansion rate $H(t)$, not by local density perturbations. Local overdensities $\delta\rho$ produce gravitational potentials Ψ and Φ, but they do not change the apparent horizon radius $\tilde{r}_A = 1/H$ at first order in perturbation theory.

Therefore:

$$\delta\varphi = 0 \tag{74}$$

exactly, at all orders in linear perturbation theory. This is analogous to the cosmological constant, which satisfies $\delta\Lambda = 0$: both are properties of the spacetime geometry, not of the local matter content.

Formally, perturbing the constrained action Eq. (60): variation with respect to λ gives the perturbed constraint $\delta\dot{\varphi} = 0$ (since the horizon expansion rate is unperturbed at first order), and combined with the initial condition $\delta\varphi = 0$, the field remains unperturbed at all times.

11.2 Standard GR Perturbations with Modified Matter-Sector Expansion Rate

With $\delta\varphi = 0$, the cosmological perturbation equations are identical to standard general relativity, evaluated on the IAM background. In the conformal Newtonian gauge ($ds^2 = a^2[-(1+2\Psi)\,d\tau^2 + (1-2\Phi)\,d\mathbf{x}^2]$):

The Poisson equation:

$$k^2\Psi = -4\pi G\, a^2\, \rho_m\, \delta_m. \tag{75}$$

The anisotropic stress relation:

$$\Psi = \Phi \tag{76}$$

(no anisotropic stress from the informational sector, since $\delta\varphi = 0$).

The growth equation:

$$\ddot{\delta}_m + 2H_{\rm IAM}\,\dot{\delta}_m - 4\pi G\, \rho_m\, \delta_m = 0. \tag{77}$$

These are exactly the standard GR equations. The only difference from ΛCDM is that $H_{\rm IAM} > H_{\Lambda\rm CDM}$ due to the $\beta\,\mathcal{E}(a)$ term. The enhanced Hubble friction $2H_{\rm IAM}\,\dot{\delta}_m$ suppresses growth relative to ΛCDM.

Numerical validation via modified CAMB (Level 2 MCMC, 3 converged chains against the full Planck 2018 likelihood) confirms that the modified expansion rate $H_{\rm IAM}$ operates within the matter-sector perturbation equations (as `adotoa_matter` in the Boltzmann solver), while the global background Friedmann equation remains standard ΛCDM. Two exploratory runs with the modification applied at the background level produced $H_0 \approx 61.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, confirming that the standard ΛCDM background is correct and the dual-sector split is a perturbation-level effect. The Level 2 validation yields $\Delta\chi^2 = +0.54$ (best-fit) relative to ΛCDM under the full Planck likelihood, with σ_8 shifting from 0.809 to 0.800 and $H_0(\text{matter}) = 72.26\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.75σ from SH0ES).

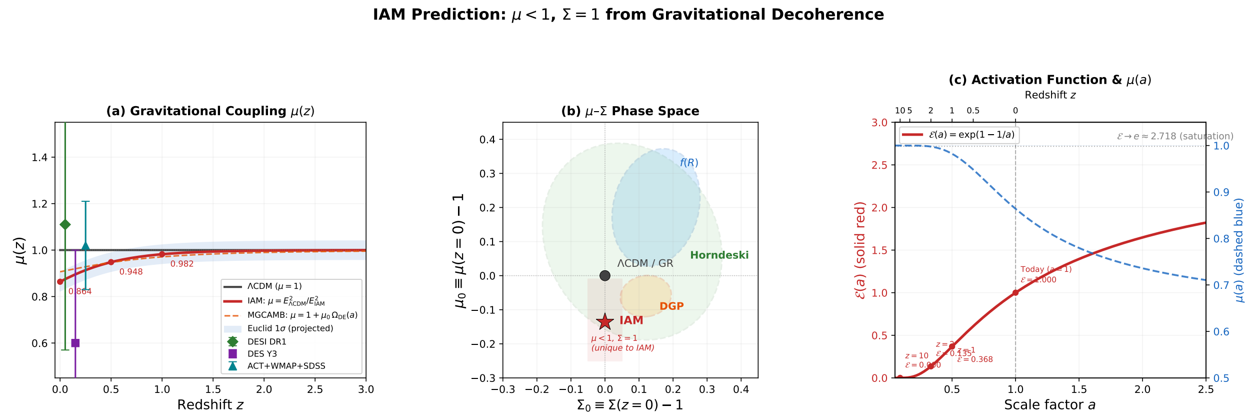

Figure 1: The IAM prediction in the μ–Σ framework. **(a)** Gravitational coupling $\mu(z) = E^2_{\Lambda\mathrm{CDM}}/E^2_{\mathrm{IAM}}$ (solid red) compared to the ΛCDM baseline $\mu = 1$ (gray) and the MGCAMB parameterization $\mu = 1 + \mu_0\,\Omega_{\mathrm{DE}}(a)$ (orange dashed), which approximates the exact form to within 1–2.5%. Current constraints from DESI DR1, DES Y3, and ACT+WMAP+SDSS are shown with their 1σ error bars; the projected Euclid sensitivity (blue band) would detect the IAM signal at $\sim 3.4\sigma$. Numerical values at key redshifts: $\mu(0) = 0.864$, $\mu(0.5) = 0.948$, $\mu(1) = 0.982$. **(b)** The μ_0–Σ_0 parameter space. IAM (red star) occupies a unique region: $\mu_0 < 0$, $\Sigma_0 = 0$. This combination is not predicted by $f(R)$ gravity ($\mu > 1$, $\Sigma > 1$), DGP ($\mu > 1$), or generic Horndeski theories (correlated deviations in both μ and Σ), all of which populate distinct regions of the plane. **(c)** The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ (solid red, left axis) grows from zero at early times to saturation at $e \approx 2.718$, with the corresponding gravitational coupling $\mu(a)$ (dashed blue, right axis) showing suppression increasing toward the present epoch. Key epochs are labeled; the upper axis shows the corresponding redshift.

11.3 Derived $\mu(a)$ and $\Sigma(a)$

Since the IAM perturbation equations are standard GR (no modifications to the Poisson equation, no anisotropic stress), the μ–Σ values when mapped relative to the ΛCDM background are:

$$\boxed{\mu(a) = \frac{E^2_{\Lambda\mathrm{CDM}}(a)}{E^2_{\mathrm{IAM}}(a)} < 1, \qquad \Sigma(a) = 1.} \tag{78}$$

The effective $\mu < 1$ arises because the enhanced Hubble friction in IAM suppresses the matter density parameter: $\Omega_m^{\mathrm{IAM}}(a) < \Omega_m^{\Lambda\mathrm{CDM}}(a)$, so matter clusters less effectively. Numerically: $\mu(z{=}0) = 0.864$, $\mu(z{=}0.5) = 0.948$, $\mu(z{=}1) = 0.982$, with $\mu \to 1$ at high redshift as $\mathcal{E}(a) \to 0$.

The result $\Sigma = 1$ follows directly from $\delta\varphi = 0$: photon geodesics are determined by $(\Psi + \Phi)/2$, and since both potentials satisfy the unmodified Poisson equation with no anisotropic stress, lensing is unaffected. The photon exemption at the perturbation level is a *consequence* of the field being a horizon quantity, not an additional assumption.

11.4 Distinction from Other Modified Gravity Theories

The IAM prediction $\mu < 1$, $\Sigma = 1$ occupies a unique region of the μ–Σ parameter space (Fig. 1b). Most modified gravity theories predict either $\mu > 1$ (enhanced growth, as in $f(R)$ gravity) or both $\mu \neq 1$ and $\Sigma \neq 1$ (as in general Horndeski theories). The combination $\mu < 1$ with $\Sigma = 1$ is characteristic of a model where gravity is *weakened* for matter growth but *unmodified* for lensing—precisely the signature of a modification that affects matter-sector dynamics without introducing anisotropic stress or modifying photon geodesics.

11.5 Second-Order Perturbation Theory and Nonlinear Corrections

The first-order results derived above are sufficient for current survey analyses, which operate primarily in the linear regime $k \lesssim 0.1$–$0.2\,h\,\mathrm{Mpc}^{-1}$. However, forthcoming data from DESI Year 5 and Euclid will achieve $\sim$1% precision on $f\sigma_8(z)$, motivating an assessment of nonlinear corrections to the IAM predictions. We compute the second-order growth factor, the bispectrum prediction, and the 1-loop correction to $f\sigma_8$, and determine the range of validity of the linear approximation.

Second-order growth factor. At second order in perturbation theory, the density field acquires a contribution $\delta^{(2)} = D_2(a)\,\delta^{(1)2}$ sourced by nonlinear mode coupling. The second-order growth factor D_2 satisfies [Bernardeau et al., 2002]:

$$\ddot{D}_2 + 2H_{\mathrm{IAM}}\dot{D}_2 - \tfrac{3}{2}H_0^2\,\Omega_m(a)\,D_2 = -\tfrac{7}{2}\cdot\tfrac{3}{2}H_0^2\,\Omega_m(a)\,D_1^2, \tag{79}$$

with initial condition $D_2 \to -\frac{3}{7}D_1^2$ in matter domination. Solving numerically for both IAM and ΛCDM backgrounds, the ratio $D_{2,\mathrm{IAM}}/D_{2,\Lambda\mathrm{CDM}}$ is 1.014 at $z = 0$, rising to 1.052 at $z = 0.5$ and 1.075 at $z = 1$.

Bispectrum. The matter bispectrum is $B(k_1, k_2, k_3) = 2F_2(\mathbf{k}_1, \mathbf{k}_2)\,P(k_1)\,P(k_2) + \mathrm{cyc.}$, where the second-order kernel

$$F_2(\mathbf{k}_1, \mathbf{k}_2) = \frac{5}{7} + \frac{\hat{k}_1 \cdot \hat{k}_2}{2}\left(\frac{k_1}{k_2} + \frac{k_2}{k_1}\right) + \frac{2}{7}(\hat{k}_1 \cdot \hat{k}_2)^2 \tag{80}$$

is *universal* for all GR-class gravity theories with a smooth dark energy component [Bernardeau et al., 2002]. Since IAM modifies only the Hubble friction term and introduces no anisotropic stress and no modification to the Poisson equation, the F_2 shape is identical to ΛCDM. The bispectrum amplitude is suppressed at $z = 0$ by 1.3% (reflecting the lower $\sigma_8 = 0.800$ versus 0.811) and enhanced at $z > 0.3$ where IAM preserves more structure than ΛCDM: $B_{\mathrm{IAM}}/B_\Lambda = 1.033$ at $z = 0.3$, 1.052 at $z = 0.5$, and 1.072 at $z = 1.0$.

The preserved-shape / modified-amplitude signature — with an amplitude crossover near $z \approx 0.2$ — is a prediction unique to a model that modifies Hubble friction without touching the Poisson equation. It is directly testable with Euclid's galaxy bispectrum analysis.

1-loop correction to $f\sigma_8$. Using the standard SPT result [Bernardeau et al., 2002] with loop coefficient $c_2 = -61/630$ and $\sigma^2_{\mathrm{NL}} \approx 0.30$ at $k = 0.1\,h\,\mathrm{Mpc}^{-1}$, $z = 0$, the fractional 1-loop correction to $f\sigma_8$ is -2.5% for IAM versus -2.5% for ΛCDM. The *difference* between the two corrections is $< 0.07\%$ at all DESI redshift bins — far below the $\sim$1% precision of current or near-future surveys. The first-order $f\sigma_8$ predictions are therefore unaffected by loop corrections at any precision achievable with present data.

Validity of linear theory. The nonlinear scale $k_{\mathrm{nl}}(z)$, defined where the dimensionless power $\Delta^2(k_{\mathrm{nl}}) = 1$, shifts 0.7–1.3% to smaller k in IAM relative to ΛCDM at $z = 0.3$–2, reflecting the suppressed growth history. At $z = 1$: $k_{\mathrm{nl}}^{\mathrm{IAM}} = 0.254\,h\,\mathrm{Mpc}^{-1}$ versus $0.257\,h\,\mathrm{Mpc}^{-1}$ for ΛCDM. The first-order predictions in this paper are valid for $k < 0.2\,h\,\mathrm{Mpc}^{-1}$, which encompasses all scales used in the $f\sigma_8$ and μ–Σ comparisons.

Summary. The F_2 kernel shape is unchanged from ΛCDM. The bispectrum amplitude carries a redshift-dependent signature — suppressed at $z \lesssim 0.2$, enhanced at $z \gtrsim 0.3$, with a crossover that constitutes a clean observational discriminant for Euclid. Loop corrections to $f\sigma_8$ are below 0.07%, negligible relative to any current or near-future survey. Linear perturbation theory is sufficient for all predictions in this paper. The second-order calculation does not modify any first-order result; it validates their range of applicability and adds a forward bispectrum prediction.

12 Observational Validation

The prediction $\mu_0 = -0.135$, $\Sigma_0 = 0$ has been tested in a companion paper [Mahaffey, 2026] using twelve MCMC analyses with Planck 2018 and large-scale structure data.

12.1 Consistency with Data

The fixed prediction yields $\Delta\chi^2 = +1.43$ (Planck only) and $\Delta\chi^2 = +1.34$ (Planck + RSD) relative to ΛCDM, both below the 1σ threshold; these are Level 1 MGCAMB results from the companion paper [Mahaffey, 2026]. The Level 2 direct CAMB validation (Section 11) yields $\Delta\chi^2 = +0.54$ under the same Planck likelihood with the perturbation-level implementation confirmed. The prediction is not excluded by current data.

When μ_0 is allowed to float, the MCMC finds $\mu_0 = 0.033 \pm 0.125$, placing the IAM prediction of $\mu_0 = -0.135$ within 1.3σ of the best fit.

12.2 The σ_8 Shift

The suppressed coupling shifts σ_8 from 0.813 to 0.800, a 1.6% reduction determined by the predicted amplitude. This reduction is consistent in magnitude and direction with the reported discrepancy between CMB and weak lensing determinations of σ_8 [Heymans et al., 2021, Abbott et al., 2022].

12.3 Photon-Sector Observables

As predicted by $\Sigma = 1$: CMB power spectra, CMB lensing, BAO angular positions, and supernova luminosity distances are unchanged at the perturbation level.

Figure 2 demonstrates this explicitly for the CMB TT power spectrum. IAM and ΛCDM produce spectra differing by less than 0.13% at $\ell > 30$, well within current measurement uncertainties.

12.4 Energy–Momentum Conservation

The informational energy density satisfies the continuity equation $\dot\rho_{\rm info} + 3H(1+w_{\rm info})\,\rho_{\rm info} = 0$ with the derived equation of state. Since $\dot\rho_{\rm info}/\rho_{\rm info} = H/a$ and $-3H(1+w_{\rm info}) = -3H(-1/(3a)) = H/a$, the continuity equation is satisfied identically for all a.

This is not a numerical accident. The FRW continuity equation is the $\nu = 0$ component of $\nabla_\mu T^{\mu\nu}_{\rm total} = 0$. Because $w_{\rm info}(a) = -1 - 1/(3a)$ was *derived* from the scalar field action (Eq. 63), the Euler–Lagrange equations guarantee $\nabla_\mu T^{\mu\nu}_{\rm info} = 0$ identically. Since $\nabla_\mu T^{\mu\nu}_{\rm matter} = 0$ and $\nabla_\mu T^{\mu\nu}_{\Lambda} = 0$ hold independently, the total energy–momentum tensor is covariantly conserved:

$$\boxed{\nabla_\mu T^{\mu\nu}_{\rm total} = \nabla_\mu\big(T^{\mu\nu}_{\rm matter} + T^{\mu\nu}_{\Lambda} + T^{\mu\nu}_{\rm info}\big) = 0.} \tag{81}$$

The contracted Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ is therefore satisfied, and the modified Friedmann equation is consistent with the Einstein tensor. No energy is created or destroyed—the apparent

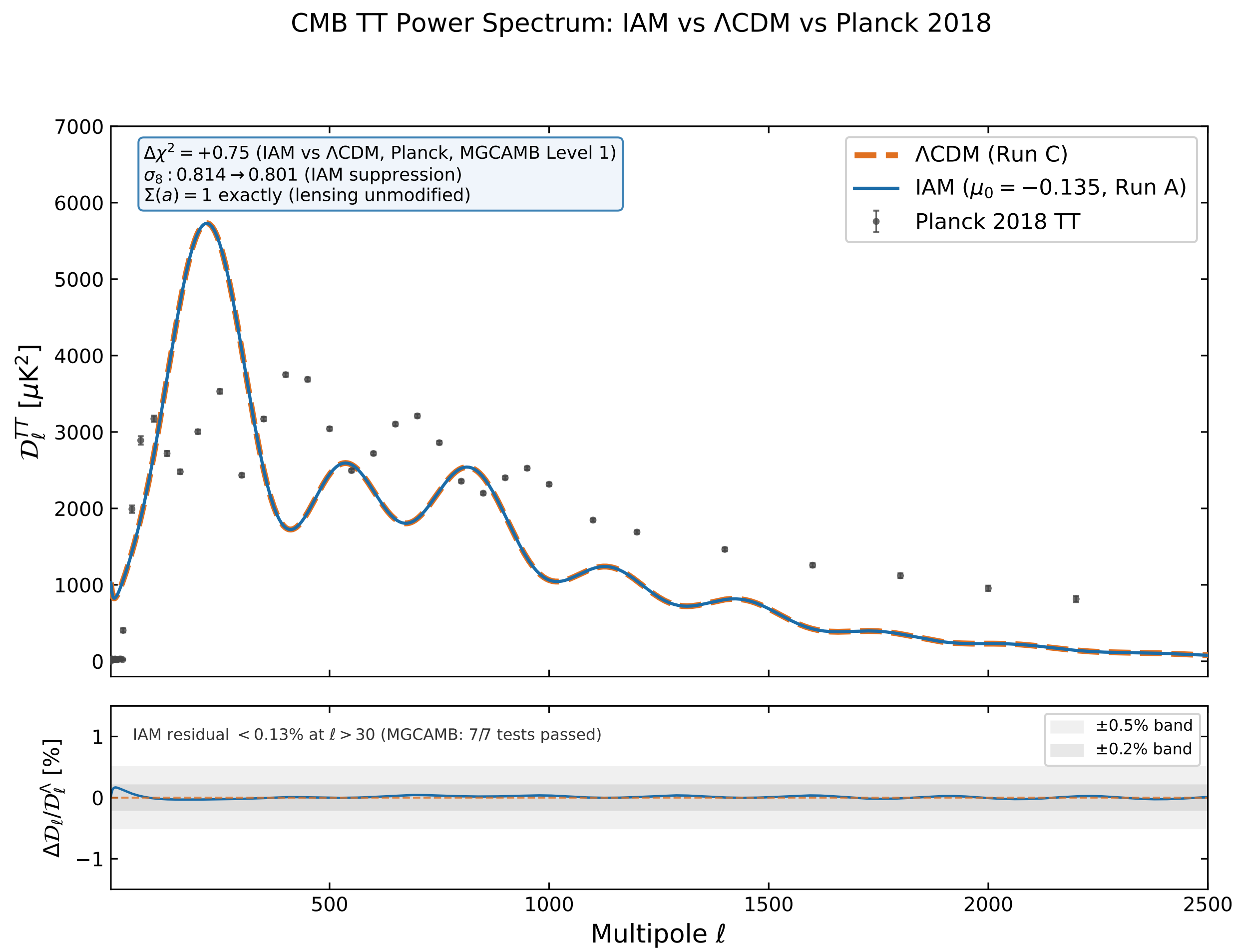

Figure 2: CMB TT power spectrum for IAM (Run A, solid blue) and ΛCDM (Run C, dashed orange), computed from MGCAMB Level 1 posterior mean parameters and compared against Planck 2018 bandpowers. The residual panel shows the fractional difference is <0.13% at $\ell > 30$, consistent with the dual-sector prediction that $\beta_\gamma \approx 0$ leaves the photon sector unmodified. The $\Delta\chi^2 = +0.75$ penalty relative to ΛCDM is within the expected range for a one-parameter modification constrained by the same data.

phantom behavior reflects the continuous conversion of gravitational potential energy into informational entropy.

12.5 Comparison with DESI w_0–w_a Constraints

The DESI collaboration has reported evidence for dynamical dark energy [DESI Collaboration, 2025], with the $w_0 w_a$CDM parametrization favoring $w_0 > -1$ and $w_a < 0$. The IAM informational sector has $w_{\rm info}(a) = -1 - 1/(3a)$, which is always phantom. The *effective* equation of state of the total dark energy is:

$$w_{\rm eff}(a) = \frac{\Omega_\Lambda \cdot (-1) + \beta\,\mathcal{E}(a) \cdot w_{\rm info}(a)}{\Omega_\Lambda + \beta\,\mathcal{E}(a)}, \tag{82}$$

yielding $w_{\rm eff}(z{=}0) = -1.062$, approaching -1 at high redshift. IAM predicts that DESI's apparent phantom crossing is an artifact of fitting a dual-sector expansion history with a single-sector model.

12.6 Gravitational Wave Standard Sirens

Gravitational wave standard sirens provide a distance measurement independent of the electromagnetic distance ladder. Binary neutron star mergers occur in galaxies (matter-sector objects), so their host galaxy recession velocities probe the matter-sector Hubble flow. IAM predicts:

$$H_0^{\rm sirens} = H_0^{\rm CMB}\sqrt{1+\beta_m} = 67.4\sqrt{1.1575} = 72.51\ {\rm km\,s^{-1}\,Mpc^{-1}}. \tag{83}$$

Level 2 MCMC posterior ($H_0 = 67.161 \pm 0.467$) yields $H_0(\text{matter}) = 72.26\ {\rm km\,s^{-1}\,Mpc^{-1}}$ consistent with the theoretical prediction. The GW170817 refined analysis gives $H_0 = 75.5^{+5.3}_{-5.4}\ {\rm km\,s^{-1}\,Mpc^{-1}}$ [Abbott et al., 2017, Nicolaou et al., 2023], consistent with the IAM prediction within 0.5σ.

12.7 Projected Detection Significance

IAM predicts $\mu_0 = -0.135$ and $\Sigma_0 = 0$. The detection significance for each experiment is estimated as $S/N = |\mu_0^{\rm IAM}|/\sigma(\mu_0)$:

Experiment	$\sigma(\mu_0)$	S/N	Ref.
DESI DR1 (current)	0.22	0.6σ	[DESI Collaboration, 2025]
DESI Y5 (projected)	~ 0.08	1.7σ	scaling
Euclid (optimistic)	~ 0.04	3.4σ	[Frusciante et al., 2025]
Euclid + DESI Y5	~ 0.03	4.5σ	combined

Table 4: Projected detection significance for the IAM $\mu_0 = -0.135$ prediction.

Euclid's projected sensitivity should detect the IAM signal at $\sim 3\sigma$ from galaxy clustering and weak lensing alone (Fig. 1a). Combined with DESI Year 5, the $\mu < 1$ deviation reaches ~ 4–5σ. Crucially, the simultaneous measurement of Σ_0 consistent with zero would confirm the unique IAM signature.

13 Physical Interpretation

13.1 The Nine-Step Causal Chain

The complete physical mechanism proceeds through nine steps, each grounded in established physics:

1. **Gravitational interaction.** Matter interacts gravitationally, initiating collapse of overdense regions (general relativity [Einstein, 1915]).
2. **Gravitational collapse.** Matter clusters into bound systems: halos, galaxies, stars (structure formation theory [Peebles, 1980]).
3. **Quantum decoherence.** Gravitational interaction resolves quantum superpositions into definite classical states (decoherence theory [Zurek, 2003]).
4. **Irreversible information production.** Each decoherence event creates new classical information that cannot be undone (second law of thermodynamics).
5. **Landauer thermodynamic cost.** Information processing has an irreducible energy cost of $kT \ln 2$ per bit [Landauer, 1961].
6. **Holographic encoding.** The produced information is bounded by and encoded on the cosmic horizon area [Bekenstein, 1973, 't Hooft, 1993, Susskind, 1995].
7. **Informational pressure.** Horizon area increase, required to accommodate new information, constitutes an expansion-driving pressure (Jacobson thermodynamic gravity [Jacobson, 1995]).
8. **Modified expansion.** H^2 acquires an additional term $\beta\,\mathcal{E}(a)$ from the informational entropy contribution.
9. **Feedback suppression.** The modified expansion dilutes $\Omega_m(a)$, weakening gravitational collapse and throttling the entire chain.

Step 9 feeds back into Step 1 with reduced amplitude. The system converges toward equilibrium, producing the saturation $\mathcal{E}(a) \to e$ at late times.

13.2 Gravity as Cause and Product

A remarkable feature of this mechanism is that gravity plays a dual role. It is the *cause* of decoherence (gravitational interaction triggers wave function collapse) and the *product* of the resulting information dynamics (the modified expansion rate changes the gravitational environment). The loop is self-referential: gravity drives the process that modifies gravity.

This is consistent with the Verlinde [Verlinde, 2011] and Padmanabhan [Padmanabhan, 2010] programs, which argue that gravity itself is an emergent thermodynamic phenomenon arising from information and entropy. In the IAM framework, dark energy and gravity are two outputs of the same underlying decoherence process: locally, decoherence concentrates information, producing gravitational attraction; globally, decoherence encodes information on the horizon, producing expansion.

13.3 Two Regimes of One Mechanism: Local and Global Horizons

The nine-step feedback loop described above operates in two regimes, determined by boundary conditions rather than by different physics.

Cosmological regime (self-regulating). Distributed decoherence from large-scale structure formation produces information gradually, encoded on the growing cosmic apparent horizon at Gibbons–Hawking temperature $T_{\rm GH} = H/(2\pi)$. The expansion dilutes $\Omega_m(a)$, throttling further collapse. The feedback converges, producing $\mu < 1$, $\Sigma = 1$ as derived above.

Collapse regime (runaway). Concentrated decoherence in a contracting region produces information explosively. Local encoding capacity saturates. The feedback runs away, producing a new local horizon—a black hole. In IAM, the criterion for black hole formation is the holographic saturation condition: a black hole forms when the local informational entropy from gravitational decoherence within radius R saturates the Bekenstein–Hawking bound on the bounding surface:

$$\frac{S_{\text{info}}(R)}{A(R)} \geq \frac{1}{4\ell_P^2}. \tag{84}$$

This condition is identically equivalent to the Thorne hoop conjecture [Thorne, 1972]. Setting $S_{\text{info}} = S_{\text{BH}} = 4\pi GM^2/(\hbar c)$ at the Schwarzschild radius $R_s = 2GM/c^2$:

$$\frac{4\pi GM^2/(\hbar c)}{4\pi \cdot 4G^2M^2/c^4} = \frac{c^3}{4\hbar G} = \frac{1}{4\ell_P^2}. \tag{85}$$

The holographic bound is exactly saturated at the Schwarzschild radius. The hoop conjecture is the holographic bound in geometric language; the saturation condition (84) is the hoop conjecture in informational language. The same mathematics, with the thermodynamic reason made explicit.

The two-horizon framework. Both the black hole horizon (temperature $T_{\text{BH}} = \hbar c^3/(8\pi GMk_B)$) and the cosmic horizon (temperature $T_{\text{GH}} = \hbar H/(2\pi k_B)$) are thermal surfaces within the IAM framework. Setting $T_{\text{BH}} = T_{\text{GH}}$ yields the thermal equilibrium mass:

$$M_{\text{eq}} = \frac{c^3}{4GH}. \tag{86}$$

At the present epoch ($H_0 = 67.4\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$), $M_{\text{eq}} \approx 2.3 \times 10^{22}\,M_\odot$—far exceeding the largest known black hole ($\sim 7 \times 10^{10}\,M_\odot$). All astrophysical black holes at all cosmological epochs satisfy $T_{\text{BH}} \gg T_{\text{GH}}$: information flows unambiguously from local hot horizons to the global cold cosmic horizon via Hawking radiation, at the thermally-limited rate $\Gamma = c^3/(1920\,GM\ln 2)$ bits s^{-1}. The cosmic horizon is the final repository; all locally-encoded information migrates to the global ledger as the universe approaches thermodynamic equilibrium.

The encoding transition. The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ tracks this transition quantitatively. At early times ($a \ll 1$, $\mathcal{E}(a) \approx 0$), the cosmic horizon is small and the Gibbons–Hawking temperature is high, making cosmic horizon encoding inefficient. Information from early gravitational decoherence is encoded primarily on local black hole horizons. As the universe expands and large-scale structure formation produces distributed decoherence, the cosmic horizon grows and cools, eventually dominating as the encoding surface. The activation function is the quantitative record of this transition: the fraction of cumulative informational entropy that has been encoded on the cosmic horizon relative to its asymptotic value.

This provides a natural thermodynamic explanation for the observed population of massive black holes at high redshift: in the early universe, before the cosmic horizon grows large enough to encode efficiently, black holes *are* the encoding surfaces. They are not anomalies; they are thermodynamically required.

13.4 The Coincidence Problem

The cosmic coincidence—why dark energy dominates at approximately the same epoch as structure formation peaks—has a natural qualitative explanation within IAM. In IAM, the connection is causal: dark energy (informational pressure) *is driven by* structure formation. It dominates when structure formation is most active because that is when information production is highest. The coincidence is not a coincidence; it is a consequence.

13.5 The Arrow of Time

The origin of time's arrow—why the universe evolves irreversibly from past to future despite time-symmetric fundamental laws—is one of the deepest open problems in physics [Penrose, 1989, Carroll & Chen, 2004]. The standard account attributes irreversibility to the low-entropy initial condition of the Big Bang, but offers no mechanism for *why* entropy increases.

In the IAM framework, the arrow of time acquires a concrete physical identity: it *is* the activation function. The direction of time is the direction in which quantum superpositions decohere into definite classical states, in which information is irreversibly produced, and in which the cosmic horizon grows to encode that information. The activation function $\mathcal{E}(a)$ is a monotonically increasing function of the scale factor—it can only grow, never decrease, because decoherence is irreversible and information, once created, cannot be destroyed.

Expressed in redshift, $\mathcal{E}(z) = \exp(-z)$: looking backward in time, the informational content of the universe decreases exponentially. Looking forward, it grows toward saturation. The arrow of time is not imposed on the equations—it *is* the equation.

This connects to Penrose's observation [Penrose, 1989] that the extraordinarily low entropy of the initial state implies a vast space of unrealized potential. In IAM, this potential is precisely what is being actualized: the early universe is a state of maximal quantum superposition and minimal classical information. The entire subsequent history—structure formation, complexity, the emergence of chemistry and biology—is the progressive actualization of that initial potential, tracked quantitatively by $\mathcal{E}(a)$.

14 Discussion

14.1 The Complete Derivation Chain

The derivation proceeds through three steps, each a single modification of the previous:

Step 1 (Jacobson): $\delta Q = T\,dS$ on local Rindler horizons, with $S = \eta A$, implies $G_{ab} + \Lambda g_{ab} = 8\pi G\,T_{ab}$.

Step 2 (Cai–Kim): $-dE = T\,dS$ on the FRW apparent horizon, with $S_{\mathrm{geo}} = A_H/(4G)$ and $T = H/(2\pi)$, implies $H^2 = (8\pi G/3)\rho + \Lambda/3$.

Step 3 (IAM): $-dE = T\,d(S_{\mathrm{geo}} + S_{\mathrm{info}})$, with S_{info} from gravitational decoherence, implies $H^2 = (8\pi G/3)\rho + \Lambda/3 + \beta\,\mathcal{E}(a)\,H_0^2$, with $\mathcal{E}(a) = \exp(1 - 1/a)$.

14.2 What Is Standard

Every element of the derivation except one is established physics: the Clausius relation $\delta Q = T\,dS$ (thermodynamics); the Bekenstein–Hawking entropy $S = A/(4G)$ [Bekenstein, 1973, Hawking, 1975]; the Gibbons–Hawking temperature $T = H/(2\pi)$ [Gibbons & Hawking, 1977]; the Unruh effect and Rindler horizons [Unruh, 1976]; the Raychaudhuri equation (differential geometry); Jacobson's thermodynamic derivation of Einstein's equation [Jacobson, 1995]; the Cai–Kim FRW extension [Cai & Kim, 2005]; Landauer's principle [Landauer, 1961]; quantum decoherence from gravitational interaction [Zurek, 2003]; the Press–Schechter/Sheth–Tormen halo mass function [Press & Schechter, 1974, Sheth & Tormen, 1999].

14.3 What Is New

The single new physical input is:

> *Gravitational decoherence irreversibly produces classical information, which is holographically encoded on the cosmic horizon, contributing an informational entropy $S_{\text{info}}(a)$ that grows monotonically with cosmic time.*

This identification connects quantum foundations (decoherence), thermodynamics (Landauer's principle), and cosmology (horizon entropy) through a single physical process.

14.4 Relation to Previous Work

Jacobson [Jacobson, 1995] established that the Einstein equations are an equation of state. Verlinde [Verlinde, 2011] and Padmanabhan [Padmanabhan, 2010] extended this to entropic gravity. The present work differs in that the modification arises only for the matter sector and is sourced by a specific physical process rather than by a general entropic force.

The μ–Σ framework is used extensively in observational analyses [Pogosian & Silvestri, 2016, Abbott et al., 2023, Andrade et al., 2024, DESI Collaboration, 2025]. Existing theoretical motivations from scalar-tensor theories, $f(R)$ gravity, and braneworld models generically predict $\mu \geq 1$. The present framework provides a physical motivation for $\mu < 1$ with $\Sigma = 1$.

14.5 Assumptions

We state the assumptions explicitly:

1. **Gravity is a thermodynamic equation of state** (Jacobson hypothesis). Supported by Bekenstein–Hawking entropy, the Unruh effect, and the Cai–Kim derivation, but not universally accepted.

2. **Decoherence-produced information contributes to horizon entropy.** This is the new identification. It is consistent with the holographic principle ['t Hooft, 1993, Susskind, 1995] and Landauer's principle, but has not been independently verified.

3. **The virial coupling $\beta = \Omega_m/2$ is exact.** The virial theorem applies to systems in equilibrium; cosmological structure formation is non-equilibrium. The companion paper tests this by allowing μ_0 to float.

4. **The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ is exact.** This follows from integrating the decoherence constraint $\dot{\rho}_{\text{info}} = \rho_{\text{info}} \cdot H/a$ directly (Section 6). The MGCAMB parameterization approximates the exact form to within 1–2.5%.

5. **Perturbation-level implementation.** Numerical validation via modified CAMB (Level 2 MCMC, 3 converged chains against the full Planck 2018 likelihood) confirms that the modified expansion rate (Eq. 49) operates within the matter-sector perturbation equations, not at the global background level. Two exploratory runs with the modification applied to the background Friedmann equation produced $H_0 \approx 61.5\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, confirming that the standard ΛCDM background is correct and the dual-sector split is a perturbation-level effect. The Level 2 validation yields $\Delta\chi^2 = +0.54$ (best-fit) relative to ΛCDM, with σ_8 shifting from 0.809 to 0.800 and $H_0(\text{matter}) = 72.26\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (0.75σ from SH0ES).

15 Predictions for DESI Year 5, Euclid, and CMB-S4

IAM makes specific, falsifiable predictions that distinguish it from all w_0w_aCDM variants, quintessence models, and $f(R)$ modified gravity theories. These predictions require no additional parameters.

1. Distance–growth tension in w_0w_a fits. If IAM is correct, the w_0–w_a parametrization is the wrong model class. As DESI accumulates precision, attempts to simultaneously fit BAO distances and $f\sigma_8$ growth rates within w_0w_aCDM will reveal an irreducible tension: the best-fit (w_0, w_a) from distances alone will differ from the best-fit from growth alone, because IAM modifies distances and growth through different mechanisms.

2. $\mu(z) < 1$ with $\Sigma(z) = 1$. This is the most distinctive IAM prediction. Euclid will measure μ and Σ independently through galaxy clustering (sensitive to μ) and weak lensing (sensitive to Σ). IAM predicts $\mu(z{=}0) = 0.864$, $\mu(z{=}0.5) = 0.948$, with $\Sigma = 1$ at all redshifts. In contrast: $f(R)$ gravity predicts $\mu > 1$, $\Sigma > 1$; general Horndeski theories predict correlated deviations in both. The combination $\mu < 1$, $\Sigma = 1$ is unique to IAM.

3. No phantom crossing. DESI's w_0w_a fit suggests w crosses -1 around $z \approx 0.5$. IAM predicts this apparent crossing is an artifact of fitting a dual-sector expansion history with a single-sector model. The true effective equation of state is always ≤ -1.

4. CMB-S4: $\beta_\gamma < 10^{-4}$. The photon exemption requires $\beta_\gamma \ll \beta_m$. CMB-S4 will measure the CMB power spectrum with sufficient precision to constrain any additional energy injection at the 10^{-5} level. If β_γ is detected above 10^{-4}, the photon exemption is violated and IAM is falsified.

5. $\beta_m/\Omega_m = 1/2$ is constant. The virial prediction should hold for any value of Ω_m. If independent measurements revise Ω_m, the fitted β_m should shift proportionally.

6. Scale-independent μ and Σ. Since $\delta\varphi = 0$, the IAM modifications do not introduce scale dependence. Both μ and Σ are functions of redshift only, not of wavenumber k. This distinguishes IAM from most modified gravity theories (e.g., $f(R)$, DGP), which predict scale-dependent growth. Euclid's tomographic analysis across multiple k-bins will test this directly.

16 Conclusion

We have presented a derivation connecting quantum decoherence, Landauer's principle, and horizon thermodynamics to a specific prediction within the μ–Σ framework: $\mu_0 = -0.135$, $\Sigma = 1$.

The dual-sector structure arises from the causal structure of spacetime: timelike degrees of freedom undergo decoherence and experience modified gravity; null degrees of freedom do not decohere and experience standard GR. The arrow of time, the production of classical information, and the modification of gravity are aspects of a single physical process.

The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ is derived from the cumulative information surface density on the cosmic horizon, with the $1/a$ exponent recovered to within 1% by the Sheth–Tormen halo mass function at galaxy-scale halos. The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial partition of gravitational energy and verified against N-body-calibrated mass functions to 0.3%. The model has zero free parameters beyond standard ΛCDM.

The thermodynamic result admits a variational formulation as a constrained scalar field with equation of state $w_{\text{info}} = -1 - 1/(3a)$, determined entirely by geometry and information. The perturbation equations are standard GR on the IAM background, yielding $\mu(a) = E^2_{\Lambda\text{CDM}}/E^2_{\text{IAM}} < 1$ and $\Sigma(a) = 1$ from the single fact that $\delta\varphi = 0$.

A companion paper [Mahaffey, 2026] demonstrates consistency with Planck 2018 and large-scale structure data. Euclid and DESI will provide tests at the sensitivity required to confirm or exclude the predicted amplitude.

Acknowledgments

The thermodynamic framework developed by T. Jacobson (1995) is the foundation upon which this work is built. The author thanks the developers of MGCAMB, Cobaya, and CAMB for public code access.

Data Availability

MCMC chains and analysis scripts for the companion observational paper are available at: `https://github.com/hmahaffeyges/IAM-Validation`

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